Effects of wrist position and support surface location on muscle activation during push-up exercise

Min-Bong, Kang1 ; Ki-Hyun, Kim2 1 Department of Physical Therapy, Daegu Medical Foundation K Hospital, Daegu, Republic of Korea2 Department of Physical Therapy, Kyungwoon University, Gumi, Republic of Korea . Min-Bong, Kang; Ki-Hyun, Kim.

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ABSTRACT (ENGLISH)

Background

The push-up is a widely used closed kinetic chain exercise that promotes shoulder joint stability through muscle cocontraction. However, how wrist and hand configurations modulate activation of key scapulothoracic and glenohumeral stabilizers (e.g., serratus anterior) remains insufficiently characterized in healthy adults, limiting mechanism-based exercise prescription.

Objective

This study examined how variations in wrist rotation, hand position, and humeral rotation during standardized push-ups affect upper-extremity muscle activation. Electromyographic (EMG) activity of the lower trapezius, upper trapezius, biceps brachii, triceps brachii, pectoralis major, and serratus anterior was measured to assess activation patterns. Findings may inform exercise prescription and rehabilitation using modified push-ups.

Methods

Thirty-one healthy young male participants (23.65±0.49 years) performed push-ups under 11 conditions combining hand position (medial, neutral, lateral), forearm rotation (pronation, supination), and shoulder rotation (internal, external). Surface EMG recorded the activity of six muscles, normalized to maximum voluntary isometric contraction (%MVIC). Group differences were tested with one-way repeated-measures ANOVA with sphericity assessment and Huynh–Feldt correction; Bonferroni-adjusted post-hoc tests and partial η^2 were reported (α=0.05)

Results

Repeated measures ANOVA showed significant effects of hand position on the lower trapezius ($p=0.01$, $\eta^2=0.088$), upper trapezius ($p=0.002$, $\eta^2=0.13$), biceps brachii ($p=0.003$, $\eta^2=0.132$), and serratus anterior ($p<0.001$, $\eta^2=0.215$). Serratus anterior exhibited the greatest activation with lateral hand placement. No significant differences were found in the triceps brachii and pectoralis major.

Conclusions

Adjusting hand and wrist positions during push-ups can selectively activate the serratus anterior, offering practical implications for rehabilitation and conditioning programs targeting scapular stability.

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